

Algebra II Functions and Logs Practice

Instructions

Complete in 60 minutes. Show clear work. Use exact values unless a decimal approximation is explicitly requested. When a question asks for a reason, give a short mathematical explanation, not just a yes or no.

Reference Facts

- A relation is a function if each input has exactly one output. A graph represents a function if it passes the vertical line test.
- A function is one-to-one if each output comes from exactly one input. A graph represents a one-to-one function if it passes the horizontal line test.
- Even and odd tests:

$$f(-x) = f(x) \quad \text{even}, \quad f(-x) = -f(x) \quad \text{odd}$$

- Inverse functions undo each other:

$$f^{-1}(f(x)) = x, \quad f(f^{-1}(x)) = x$$

- Log properties:

$$\log_b(MN) = \log_b M + \log_b N, \quad \log_b\left(\frac{M}{N}\right) = \log_b M - \log_b N, \quad \log_b(M^r) = r \log_b M$$

$$\log_b(b^x) = x, \quad b^{\log_b x} = x, \quad \log_b x = \frac{\log_a x}{\log_a b}$$

Problems

1. The relation contains the ordered pairs

$$(-5, 2), (-3, -1), (-1, 2), (1, 6), (3, 10), (5, 6).$$

Determine whether the relation is a function. Then determine whether its inverse relation is a function. If the inverse relation is not a function, give the smallest possible restriction of the original domain that keeps exactly four points and makes the inverse a function.

2. Let

$$f(x) = \frac{2x - 7}{x + 4}.$$

Find the domain of f , the range of f , the x - and y -intercepts, and both asymptotes.

3. Determine whether each function is even, odd, or neither. Justify algebraically.

$$f(x) = x^6 - 4x^4 + x^2 - 9, \quad g(x) = \frac{x^5 - 3x}{x^2 + 1}, \quad h(x) = \frac{x^3 - 2x + 5}{x^2 + 4}.$$

4. A function p is given by

$$p(x) = \begin{cases} -2x + 1, & x < -1 \\ x^2 - 4, & -1 \leq x < 3 \\ 2x - 13, & x \geq 3 \end{cases}$$

Find $p(-4)$, $p(-1)$, $p(3)$, and every x such that $p(x) = 0$. Then state whether p is one-to-one on its full domain.

5. Let

$$f(x) = \frac{x - 1}{x + 2}, \quad g(x) = x^2 - 4.$$

Find $(f \circ g)(x)$ and its domain. Then find $(g \circ f)(x)$ and its domain.

6. Find the inverse of

$$f(x) = \frac{5x + 6}{3x - 2}.$$

State the domain of f and the domain of f^{-1} . Then verify algebraically that $f(f^{-1}(x)) = x$.

7. The graph of $y = f(x)$ passes through the points

$$(-4, 7), (-2, 1), (0, -3), (3, 1), (6, 9).$$

List the corresponding points on the graph of $y = f^{-1}(x)$. Then determine whether the given points could all lie on a one-to-one function. Explain geometrically.

8. Let

$$q(x) = (x - 2)^2 - 5.$$

Restrict the domain of q so that it has an inverse function and the restriction includes $x = 2$. Find the inverse for that restricted function, and state its domain.

9. Expand completely:

$$\log_b \left(\frac{27x^5 \sqrt[3]{y^4}}{b^2 \sqrt{z}(x-1)^3} \right).$$

10. Condense into a single logarithm:

$$\frac{3}{2} \log_3(x-1) - \frac{1}{2} \log_3(y+2) + 2 \log_3(9) - 3 \log_3(z).$$

11. Solve and check for extraneous solutions:

$$\log_2(x-1) + \log_2(x-5) = 3 + \log_2(x+3).$$

12. Solve. State the domain restriction used.

$$\log_4(2x+7) - \log_4(x-1) = \log_4(3).$$